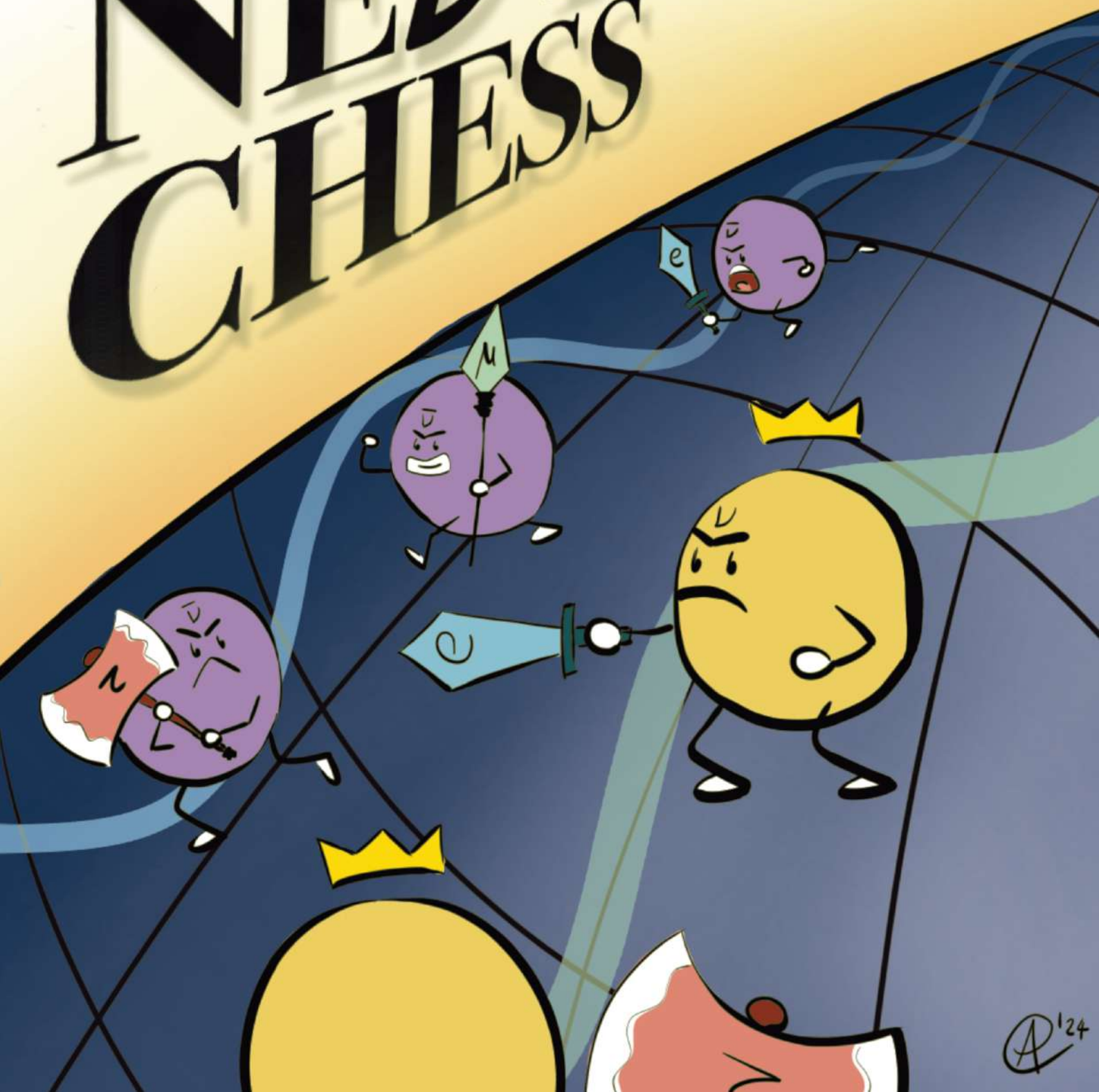


Andrea Ciarma

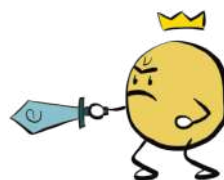


NEUTRINO CHESS





NEUTRINO CHESS



2 Players - 15-30 Minutes
Concept and illustrations by Andrea Ciarma

Command your Neutrino army in a galactic strategic challenge! Remember that Neutrinos change flavor as they move according to the Neutrino Oscillation phenomenon. Plan your moves carefully to capture enemy pieces and protect your own.

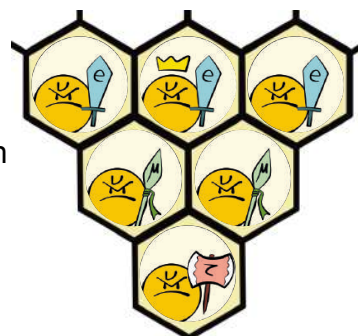
Object of the game: Checkmate the opponent's Neutrino King, or capture all other Neutrinos.

Components: 1 Board, 1 Manual, 10 Neutrino Pieces, 2 King Neutrino Pieces

Preparation

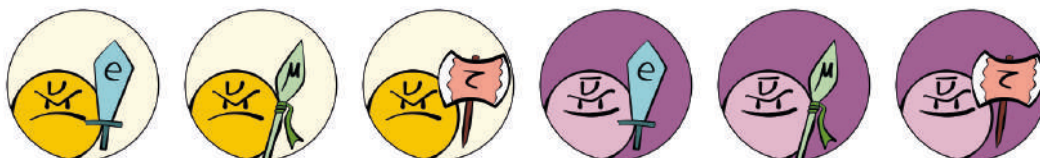
Place the board on the table. One Player will drive the **Neutrinos** ν (yellow color) and the other will drive the **Anti-Neutrinos** $\bar{\nu}$ (purple color). Flip a coin to decide the **First Player**.

For the first few games, each player follows the **initial setup** suggested in the figure. For games between experienced players, players can alternate placing a token of their choice in their deployment zone, taking care to place the Neutrino King in the King Cell.

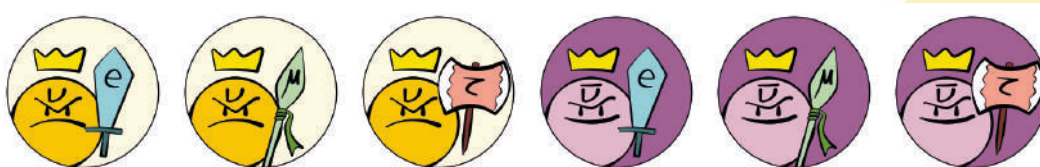


Definizioni

Neutrino Pieces - represent the two armies of Neutrinos ν and Anti-Neutrinos $\bar{\nu}$. Each Neutrino Piece can exist in 3 **Flavors**: electron e , muon μ and tau τ .

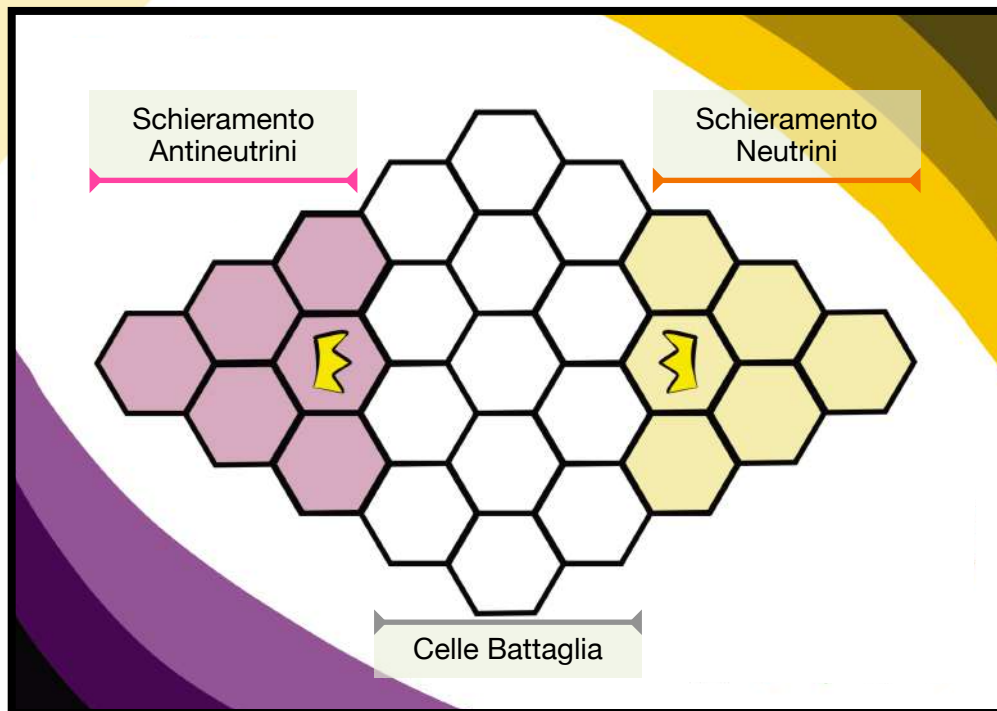


King Neutrino Pieces - they are the commanders of the two armies, the most important Pieces! They are recognized by the Crown . Like Neutrino Pieces, they come in 3 different Flavors.



Cells - The board is made up of 3 types of cells:

King Cells  , Deploy Cells   and Battle Cells .



How to play

Neutrino Chess is a turn-based game. Starting with the First Player, each player may take one of the following actions: **Move** or **Deploy a Captured Piece**.

Move - you can move one of your Pieces **1 or 2 Cells** on the Board. Due to the Oscillation phenomenon, your **Neutrino Piece will change Flavor** depending on the distance it travels:

1 Cell (Short Range)   

2 Cells (Long Range)   *a scelta

To change Flavor, flip the Neutrino Piece to show the correct face.

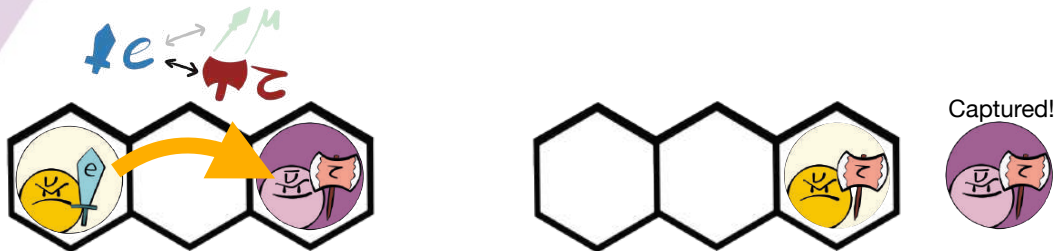
- A Piece in a King Cell or Placement Cell can only be moved forward, and from a Battle Cell it cannot move to a King or Deployment Cell.



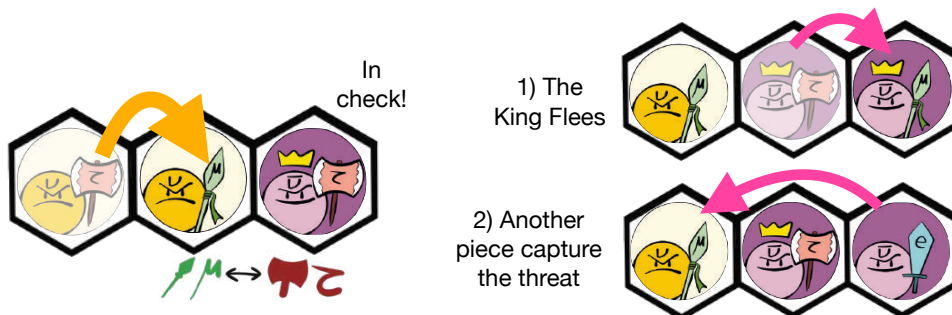
- For the movement of 2 Cells, it does not matter whether the middle cell is occupied or not.



- You can only move to an empty cell or a cell occupied by an opponent's piece of the same flavor as your piece at the end of your move. In the latter case, you **capture** the opponent's piece. Remove it from the board and place it next to your side of the board (see Placing a Captured Piece).



- If a Neutrino King could be Captured after a move, it is said to be **In Check** and must be moved to a safe location or the threat removed. **Warning: Neutrino Kings cannot Capture Neutrino Pieces!** Neutrino Kings can be put in check even when they are on the King Cell. If there is no way to save the King, **Checkmate** is declared and the game ends (see End Game).



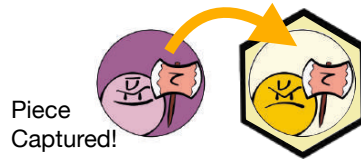
Moving your Neutrino King into a cell where it would otherwise be in check is an **illegal move**.



Neutrino Kings can only "capture" each other's Neutrino Kings. This means that moving your Neutrino King into a Cell where it could be captured by your opponent's Neutrino King is an **illegal move**.



Deploy a Captured Piece - Place a captured piece into any of your Deployment Cells under your control (changing its type $\nu/\bar{\nu}$ or depending on your team). You cannot change the piece's flavor before deploying it. From now on, treat it like any other piece of yours.



End of Game

The game ends if one player checkmates the other (the opponent's Neutrino King is in check, cannot be moved to a safe cell, or the piece that checkmates cannot be captured on the next turn), or if a player is left with only his Neutrino King.

Neutrinos and Flavor Oscillation

Neutrinos are fundamental particles, just like electrons, photons, and quarks, meaning they cannot be divided into smaller parts. Their name already tells us something important: neutrinos are incredibly light—in fact, they are the lightest known massive particles (although we still do not know their exact mass!)—and they carry no electric charge.

Neutrinos interact only very weakly with matter, so much so that they can pass straight through the entire Earth without any difficulty! This makes them extremely difficult to detect. To observe even a few of them, we need enormous detectors capable of "watching" very intense sources of neutrinos.

The first neutrinos we discovered were **electron neutrinos**, which are typically produced in radioactive decays and nuclear processes. They are called electron neutrinos because they are most closely associated with the electron, the charged particle with which they interact most often. There are also two other types of neutrinos: **muon neutrinos** and **tau neutrinos**, named after the electron's two heavier cousins, the muon and the tau. We refer to these different types as **flavors**—although, of course, no one has ever tasted a neutrino!

However, physicists eventually realized that neutrinos behave in a very peculiar way. A neutrino is born with a specific flavor, but as it travels, it does not necessarily remain the same...

The First Clues: Homestake and Kamiokande

In the 1970s, the **Homestake** experiment in the United States observed electron neutrinos coming from the largest nuclear reactor we have: the Sun! Every second, about **10 billion solar neutrinos pass through each square centimeter of the Earth's surface**. The detector consisted of a large tank filled with **380,000 liters of tetrachloroethylene** (the same chemical commonly used in dry cleaning), which was capable of detecting electron neutrinos.

The results, however, were astonishing: the experiment detected only about **30% of the expected electron neutrinos**. It seemed that something was wrong—either with the theoretical predictions or with the experiment itself.

The solution to this puzzle was proposed by **Bruno Pontecorvo**. He suggested that the electron neutrinos produced in the Sun could change their flavor as they traveled through space toward the Earth. In other words, electron neutrinos left the Sun, but during their journey they transformed into muon or tau neutrinos, which the Homestake detector was unable to observe.

A few years later, the **Sudbury Neutrino Observatory (SNO)**, which was capable of detecting **all three neutrino flavors**, confirmed the theoretical prediction by measuring the expected total number of solar neutrinos.

Short and Long Range Oscillations

Neutrino oscillations can occur over both **short** and **long distances**, depending on the neutrinos' energy and how far they have traveled from the point where they were produced.

- **Short-baseline oscillations:** These are studied in experiments where neutrinos are produced in the laboratory and detected after traveling only a few kilometers. One example is the **OPERA** experiment at the **Gran Sasso National Laboratory** in Italy, which observed muon neutrinos produced by an accelerator at **CERN** in Geneva and searched for those that had transformed into tau neutrinos.
- **Long-baseline oscillations:** These occur over much greater distances, such as between the Sun and the Earth or in atmospheric neutrinos produced when cosmic rays strike the Earth's atmosphere. Experiments such as **Super-Kamiokande**, **SNO** in Canada, and **KM3NeT** in the Mediterranean Sea investigate this phenomenon.

In the game, the way the **Neutrino Pieces** change flavor as they move across the board is inspired by this very phenomenon. The pattern of flavor changes is not random: it reflects the real oscillation probabilities of different types of neutrinos observed in nature.

How does it work?

The phenomenon of **neutrino oscillation** is linked to one of the most intriguing features of quantum physics. As we have seen, neutrinos come in three **flavors**: electron, muon, and tau. When they are produced—for example, in the Sun or inside a nuclear reactor—they begin their journey with one of these flavors.

But this is where things become more complicated: neutrinos do not travel through space as flavor states. Instead, they propagate as **mass states**, usually denoted ν_1 , ν_2 , and ν_3 . These mass states behave like waves, each oscillating with a slightly different frequency. As the neutrino travels, the combination of these three mass states evolves over time, so that the neutrino may correspond to a different flavor than the one it started with. In other words, a neutrino that is produced as an electron neutrino can later be detected as a muon or tau neutrino, and vice versa. This continuous change of flavor is what we call **neutrino oscillation**.

The discovery of neutrino oscillations was revolutionary. It showed that neutrinos must have mass—even if it is incredibly small—and opened a new window onto the fundamental laws of the Universe. So, although neutrinos are almost invisible and incredibly elusive, they have an extraordinary story to tell about the smallest building blocks of nature.

